LAB EXPERIMENT

COURSE CODE: ITA0318

COURSE NAME: MOBILE COMPUTING

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EXP -15

15) Design an interactive presentation in Figma explaining the issues in mobile IP.

Aim:

To Design an interactive presentation in Figma explaining the issues in mobile IP

Procedure:

- Define Presentation Structure
- Create a Figma Project
- Design Visual Elements
- Make it Interactive
- Add Annotations and Explanations
- Incorporate Multimedia
- Storyboard Animation
- Test the Prototype
- Collaborate and Gather Feedback
- Finalize and Share

Design:

SET 15 / NETWORK SYSTEMS

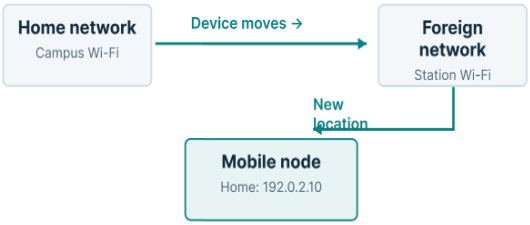
ISSUES IN MOBILE IP

Issues in Mobile IP

How can a device change networks without losing the address others use to reach it?

THE CORE IDEA / A STABLE IDENTITY, A MOVING LOCATION

Keep a home address. Update where packets should go.



A phone leaves campus for the station. Its care-of address changes; its home address remains the reachability anchor.

Explore the issues

Static lesson map · detailed slides below

SLIDE 03 ↗
Address changes

SLIDE 04 ↗
Registration & delivery

SLIDE 05 ↗
Triangular routing

SLIDE 06 ↗
Handover & packet loss

SLIDE 07 ↗
Security threats

SLIDE 08 ↗
Overhead & compatibility

Start with the architecture on 02. Compare improvements on 09.

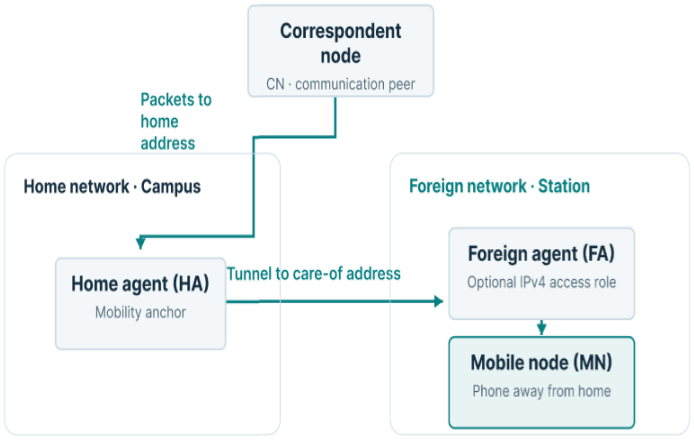
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ISSUES IN MOBILE IP

One identity. Several network roles.

The home agent keeps track of a mobile node that is currently away from its home network.

BASIC MOBILE IPv4 ARCHITECTURE



Home agent

Receives traffic for the home address and tunnels it toward the node's registered care-of address.

Foreign agent: optional

Mobile IPv4 may use an FA care-of address or a co-located CoA on the mobile node.

Mobile IPv6 distinction

No Foreign Agent is required. The mobile node configures its own care-of address.

An address is also a location.

Ordinary subnet movement changes the IP address used for routing—and can disrupt ongoing communication.

EXAMPLE / THE SAME PHONE AT TWO LOCATIONS

AT HOME / CAMPUS

192.0.2.10

Home address on home link

AWAY / STATION

198.51.100.24

Co-located care-of address



The mapping changes; the home identity does not.

Documentation-only IPv4 addresses · simplified co-located CoA example

Without mobility support

Packets still sent to the old subnet do not automatically find the device on its new network.

With Mobile IP

The home address remains stable while mobility signaling updates the current care-of address.

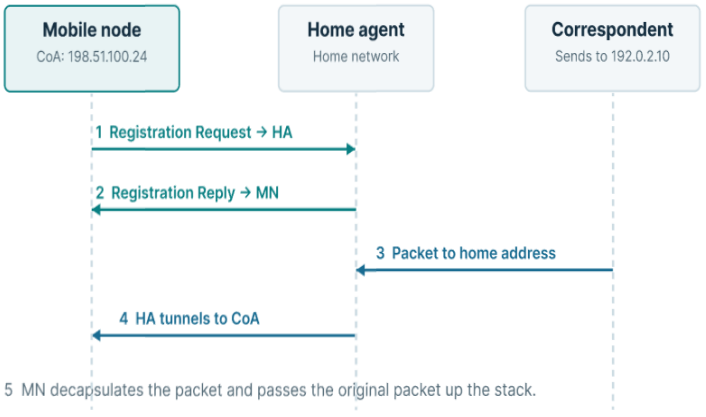
Why sessions care

An ordinary IP address change can invalidate transport endpoints. Mobility aims to preserve reachability and continuity.

Register the location. Tunnel the packet.

Control signaling establishes a binding; data forwarding uses that binding to deliver traffic to the mobile node.

MOBILE IPv4 / CO-LOCATED CoA EXAMPLE



If using a Foreign Agent

The MN sends its IPv4 registration via the FA. The HA tunnels to the FA's CoA; the FA delivers to the MN.

If using a co-located CoA

The MN registers with the HA and terminates the tunnel itself, as illustrated here.

Mobile IPv6 uses a Binding Update

The MN informs its HA of its care-of address. Do not call this Mobile IPv4 registration.

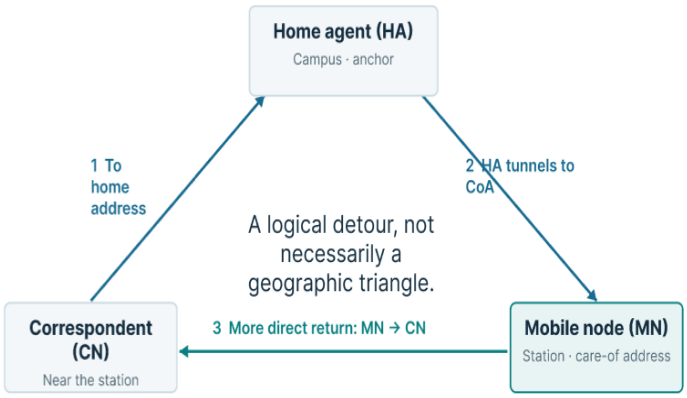
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ISSUES IN MOBILE IP

The route can take a detour.

Basic Mobile IPv4 may deliver CN → HA → MN, even when the correspondent and mobile node are close together.

TRIANGULAR ROUTING / BASIC MOBILE IPv4



Blue: inbound via HA / Teal: direct outbound path when permitted

Why it can be inefficient

The HA may be far from both endpoints. Extra path length can increase delay and anchor traffic.

Reverse tunneling changes the path

When used, MN → HA → CN sends the return traffic through the home agent too.

Mobile IPv6 route optimization

A correspondent can learn the binding and send directly to the CoA, subject to protocol and deployment support.

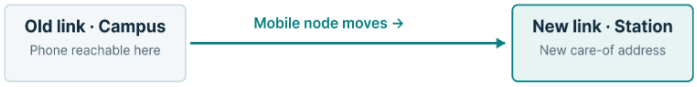
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ISSUES IN MOBILE IP

Movement creates a delivery gap.

Handover is not instantaneous: link change, address configuration, and mobility signaling all take time.

ILLUSTRATIVE TIMELINE / NOT MEASURED PERFORMANCE



0–40 ms	40–160 ms	160–240 ms
Link interruption	Address configuration	Mobility update

P1 ✓	P2 ×	P3 ×	P4 …	P5 ✓

Example only: × lost during the gap, … delayed. Real durations and buffering behavior depend on the network and implementation.

Old path becomes stale

Packets may continue toward the previous location until the new binding is effective.

Real-time traffic notices

Voice or video can pause. Lost packets may trigger retransmission for reliable transports.

Reduce—but do not assume zero—loss

Faster signaling, link-layer coordination, and buffering can help, with complexity and deployment trade-offs.

Set 15 · 07 · A location update is a security decision.

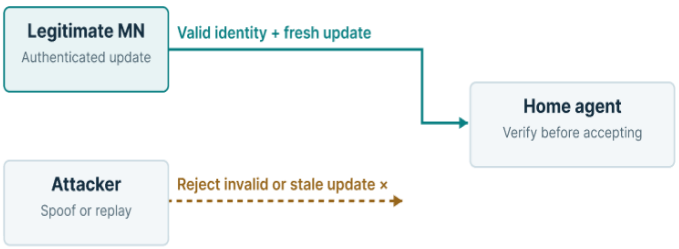
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ISSUES IN MOBILE IP

A location update is a security decision.

If a false binding is accepted, an attacker can redirect traffic. Authentication and freshness checks protect the mapping.

THREAT MODEL / WHO IS ALLOWED TO CHANGE THE BINDING?



Spoof / redirect

False CoA sends traffic to the wrong place.

Replay

An old valid update can restore a stale location.

Mobile IPv4

Authenticate registration between MN and HA. Identification fields support replay protection using timestamps or nonces.

Mobile IPv6

Protect home-agent Binding Updates with IPsec. Correspondent-node updates use binding-management protections.

Return routability has a scope

It checks reachability at home and care-of paths for route optimization. It does not replace strong authentication in every threat model.

Set 15 · 08 · Mobility adds work to the network.

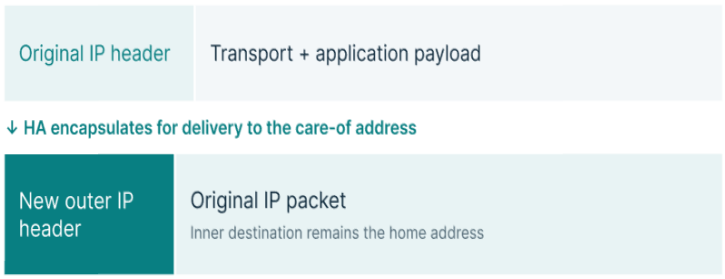
SET 15 / NETWORK SYSTEMS

ISSUES IN MOBILE IP

Mobility adds work to the network.

Encapsulation, mobility state, and middleboxes create practical deployment constraints—not just routing questions.

TUNNEL OVERHEAD / A PACKET INSIDE A PACKET



Less room for the original packet

For basic IPv4-in-IPv4, the outer IPv4 header adds at least 20 bytes. Account for tunnel MTU and Path MTU Discovery to avoid fragmentation or dropped oversized packets.

Encapsulation cost

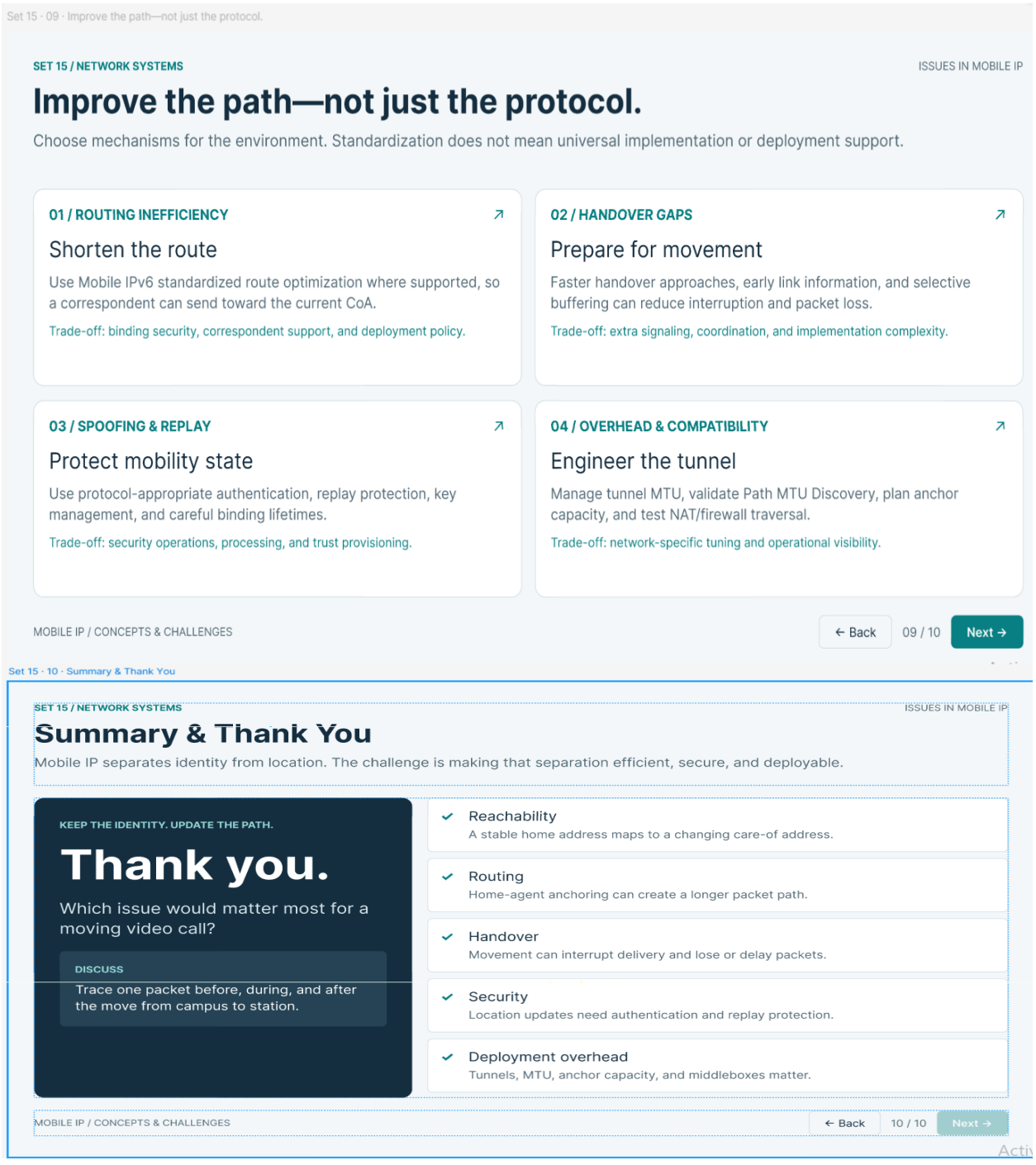
Header size depends on tunnel type and options. Added bytes consume bandwidth and reduce effective payload capacity.

Anchor load & state

The HA maintains bindings and forwards traffic for many nodes. Capacity and resilience become design concerns.

NAT & firewall compatibility

Especially in IPv4, middleboxes can block tunnel protocols or mobility signaling. Traversal support must match the deployment.



Result: -

Hence the interactive presentation in Figma explaining the issues in mobile IP is created and executed successfully.